

ZetaPay – Payments Management System

Project Documentation & Logic Walkthrough

1. Logic Applied to the Project

- **Centralized State Management (Vuex) :**
The application uses Vuex as a single source of truth for users and payments data. Whenever a payment is created or updated, the change is automatically reflected across the Dashboard, Payment List, and User views. This keeps data consistent throughout the app.
 - **Traceability Logic :**
To meet fintech traceability requirements, automated timestamping is implemented inside Vuex actions.
Each record maintains `createdAt` and `updatedAt` fields, which are updated by system logic whenever data is created or modified. This provides a reliable audit trail.
 - **Conditional Form Routing :**
The `PaymentForm.vue` component detects whether it is in **Create** or **Edit** mode using Vue Router hooks.
This allows the same form component to be reused while conditionally displaying fields like **Status**, which appear only during the update flow.
 - **Input Validation & Data Integrity :**
Strict numeric range validation is applied to payment amounts.
TypeScript enums are used for **Category** and **Status**, ensuring only valid values are allowed and improving overall UI and data robustness.
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2. Testing Methodology

Three unit test suites were implemented using **Jest** and **Vue Test Utils**:

- **PaymentValidation.spec.ts :**
Validates core business logic. Ensures the *Confirm Transaction* button is disabled when the payment amount is zero or negative.
 - **PaymentForm.spec.ts :**
Verifies that the payment form mounts correctly and interacts with Vue Router using mocked `useRoute` and `useRouter`.
 - **PaymentUI.spec.ts :**
Confirms correct UI rendering from the Vuex store, including validation that the user dropdown displays the correct number of users.
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3. Instructions to run the project :

Follow these steps to run and verify the project locally :

1. Extract the provided ZIP folder.
2. Navigate to the project root : `cd payments-management-system`
3. Install dependencies : `npm install --legacy-peer-deps`

4. Launch the application : `npm run dev`
 5. Run unit tests : `npm run test:unit`
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4. GitHub Repository

GitHub Link:

<https://github.com/HimaVarshaKaranam/payments-management-system>